

SPEED WIND PREDICTION USING AI MODELS, A COMPARISON OF TWO ALGORITHMS USING LOCAL COLOMBIAN DATA.

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ABSTRACT

In Colombia, there are territories disconnected from the national electrical network, all of which belong to the off-grid zone. Main and secondary services are not guaranteed in these areas, resulting in the unavailability of electricity, water, and healthcare, among others. This problem also causes slower development in these communities compared to other areas where an electrical network is already in place. Renewable energy sources are an interesting alternative in these territories as they have solar and wind resources that can be used to generate constant electricity. Using these two options, which are the most common and make them less expensive and easier to develop, can be a valuable way to reduce gaps in existing services. Based on the presented information, the Universidad Distrital Francisco José de Caldas has installed five monitoring stations in the Cundinamarca region to evaluate current conditions and assess possible electrical potential. These monitoring stations can measure various features, including solar radiation, wind speed, wind direction, temperature, and humidity. They are also located in different places, with varying climates because these cities have different elevations, the lowest at 700 m.a.s.l. medium at 2500 m.a.s.l and higher at 2800 m.a.s.l. Based on the available big data and the high-frequency rate (every 5 minutes), AI techniques, including artificial neural networks (ANNs, LSTMs) and random forests (RFs), are used to predict future wind speed values. This prediction seeks to identify the test data and evaluate them using metrics such as MSE, MAE, and the coefficient of determination (R^2). In this study, the expected result is to reduce the error, e.g., the MAE value of 1.0736 obtained in the LSTM after varying the number of epochs and the activation function.

Keywords: Artificial Intelligence, Machine learning, Renewable sources, Wind speed.

NOMENCLATURE

ANN: Artificial Neural Network

M.A.S.L: Meters Above Sea Level

LSTM: Long Short-Term Memory

MSTL: Multiple Seasonal-Trend decomposition using LOESS

MAE: Mean Absolute Error

MSE: Mean Square Error

R^2 : Determination coefficient.

INTRODUCTION

In recent years, climate change has affected the world, introducing new sources into each country's electric grid. Renewable energy becomes increasingly important every day, offering an alternative to oil-based energy dependency and generating a smaller carbon footprint. Recent studies [1]–[3] have explored new territories for the development of renewable energy. In the case of Solar and Eolic energy, it is important in desert and high-temperature regions. Additionally, research has addressed important topics, such as the risks these systems face when exposed to high temperatures or similar conditions commonly found in desert zones. In [4], how solar panels are affected by these conditions, affecting their life cycle, among other features, has been studied.

As renewable energy has become more relevant in recent years, Eolic is a new alternative for cities where solar energy is not a common option. This has been examined in [5], [6] cases where Eolic energy is a feasible option for cities such as Bogotá, Colombia, and Veracruz, Mexico. These studies indicated that its implementation is possible under local conditions. Furthermore, several works, such as [7], explain how wind energy can contribute to the electrical system when excess generation is sold to utility companies. This aspect has become

important because it can reduce the economic impact of implementing Eolic systems.

Another technology has become more important in recent years: Artificial Intelligence AI is a technology that has the capacity to reduce the time and complexity involved in solving problems. In the energy sector, AI is a powerful tool for planning and evaluating energy consumption and demand. Statistical techniques [8]–[10] have been used in combination with Machine Learning to improve energy management, which will impact the system, improving the user experience. In [11] there is a study about the use of AI to evaluate possible future energy demand in Colombia. As AI has become more important across various fields, it has been integrated in renewable energy, and [12], [13] AI has predicted solar energy, which is the most widely adopted source in recent years. In addition to solar energy, AI has been used for wind energy. Recent research has focused on predicting Wind speed, with improved accuracy achieved by incorporating additional parameters such as location, prediction time, and datasets from specific regions. Studies [14]–[20] present these results, indicating that Eolic energy is now an option for several territories. In addition to AI applications for predicting and evaluating renewable energy [21]–[23], several studies have focused on identifying the most suitable locations for renewable energy, particularly in isolated regions. These projects are focused on replacing oil-based dependency with renewable energy while addressing needs in off-grid territories where connection to the electric network is not possible. Finally, important considerations for renewable energy projects in such areas include the potential for high costs. It [24] outlines key factors that support the development of similar initiatives in isolated territories where support is necessary to develop a new system. They showed that it is essential to provide economic support and special conditions, such as low taxes, to enable vulnerable populations to adopt and sustain renewable energy.

MATERIALS

This Project was developed using a dataset from two monitoring stations belonging to the Universidad Distrital Francisco José de Caldas. In this case, the key variable is wind speed, taken from a bigger group that includes temperature, humidity, atmospheric pressure, among others. As the stations are in different locations, two were chosen. First was Anapoima station, with a warm climate and an average elevation of 710 m.a.s.l., and second was Paiba station, in Bogotá. Its height is around 2600

m.a.s.l., and it has a cold climate. The frequency rate in this case is 10 minutes, which produces 144 data points per day, and the dataset used for this project spans about 6 months. Training and test for the selected algorithms were done on a laptop using a Ryzen 5 3550H processor, 16 GB of RAM, and a 4GB graphics card, GTX 1650.

AI ALGORITHMS

The algorithms selected for this project are the Neural Network Long-Short-Term Memory (LSTM) and the Multiple Seasonal-Trend decomposition using LOESS (MSTL), which incorporate Seasonal ARIMA as the main trend forecaster, among others.

LSTM: In this case, the key variable is wind speed, using a training size of 85% of the data and the remaining for testing. In this case, the window size is changed using 10, 30, and 50 samples. It used different optimizers for the Activation function; in this case, only ReLU and Tanh were used. In this process, the target was to identify the lower error metrics and achieve the lowest training time.

MSTL: In this case, the identified temporality is 24 hours in the dataset. It produced a 144-daily dataset, and this algorithm used three Trend Forecasters to identify the best. Additionally, in this setting, the window size is adjusted, and diary or weekly behavior is evaluated to improve the prediction of real data. The target is to identify the parameter combination that produces the lowest errors.

MODELLING

Figure 1 illustrates the wind speed recorded in Bogotá (Paiba station), which shows higher values than observed at Anapoima station in Figure 2. The difference is attributed to the specific location of each station; in Bogotá, the monitoring station is not facing any obstacles, and the city's colder weather could lead to higher wind speeds. In contrast, Anapoima is at a lower altitude and in a warmer climate, where lower wind speeds have been recorded across all measurements. The prediction was performed using two AI algorithms: LSTM and MSTL.

RESULTS AND DISCUSSION

Tables 1 and 2 show the LSTM results obtained from the wind signal from the Paiba climate station. Based on the error metric, both activation functions perform similarly. However, the R^2 coefficient was always below 0.5, indicating that the predicted signal did not accurately reflect the actual wind speed data.

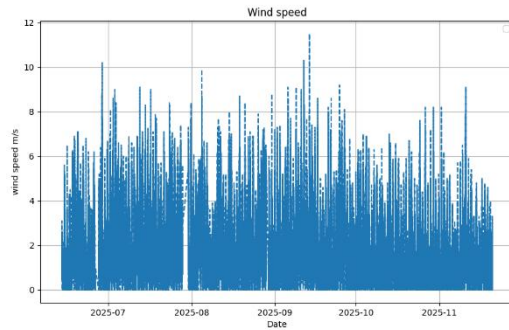


Figure 1. Wind speed signal from Paiba weather station (Bogotá)

Across all cases from Paiba station, including different window sizes and optimizer selections, the MAE metric remained around 0.7, and the MSE was approximately 0.8. Based on the result, the algorithm will require an extra adjustment to improve accuracy. In this case, these two activation functions were retained, and other options were discarded in preliminary tests. For the Anapoima station, the results have similar behavior to those reported in the Paiba station, with a low R^2 coefficient. This means the predicted values do not capture the data's variability and tend to produce an average rather than the high-value spikes observed in the actual measurements.

On the other hand, it is important to note that MAE and MSE error metrics are slower on this dataset, with all values around 0.3 and 0.6, respectively, as shown in Tables 3 and 4. This indicates that the predicted values are closer to the actual wind speed signal in Anapoima and that the results are better aligned with expectations. However, the R^2 values remain low, even lower than Paiba's prediction, suggesting that this model struggles to correctly correlate the obtained results with the original signal.

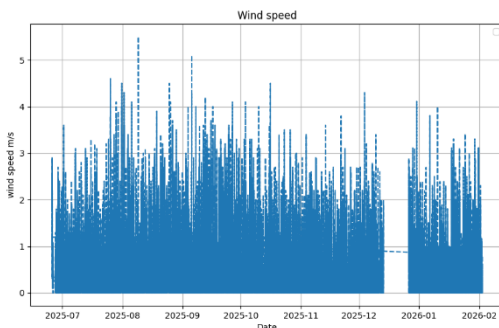


Figure 2. Wind speed signal from Anapoima weather station

Table 1. LSTM error metrics using ReLu Function, different Optimizer, and Size Window for Paiba station dataset.

ReLu		Size window		
Optimizer	Error	10	30	50
Adam	MAE	0.6803	0.7179	0.6977
	MSE	0.8264	0.8612	0.8223
	R^2	0.4498	0.4268	0.4509
Nadam	MAE	0.6808	0.6926	0.6928
	MSE	0.8090	0.8195	0.8176
	R^2	0.4614	0.4545	0.4540
Adamax	MAE	0.6869	0.6675	0.6905
	MSE	0.8161	0.8051	0.8175
	R^2	0.4566	0.4641	0.4541

Figure 3 illustrates the actual wind speed signal and zooms in on the predicted data. As shown, the predicted signal captures the general tendency of the original signal, particularly in low and average values. However, it does not capture the higher or maximum values in the predicted signal, achieving only half of the observed peak values. This underestimation could lead to energy losses in a wind energy system, as the turbines may operate below their minimum speed. Overall, while the predicted signal reflects the global behavior of the actual data, its amplitude is lower than the observed values. In the case of Anapoima the prediction error was consistently lower across all models. A likely explanation is the lower maximum wind speeds observed at this site, while peak values in Anapoima reached approximately 5.5 m/s, Paiba recorded peaks of up to 11.5 m/s. These lower values are easier for the model to replicate, resulting in a smaller error magnitude.

Table 2. LSTM error metrics using Tanh Function, different Optimizer and Size Window for Paiba station dataset.

Tanh		Size window		
Optimizer	Error	10	30	50
Adam	MAE	0.7036	0.6945	0.7050
	MSE	0.8494	0.8344	0.8374
	R^2	0.4345	0.4446	0.4409
Nadam	MAE	0.69398	0.6727	0.6727
	MSE	0.8283	0.8120	0.8056
	R^2	0.4485	0.4595	0.4621
Adamax	MAE	0.6834	0.6799	0.6776
	MSE	0.8167	0.8107	0.8108
	R^2	0.4562	0.4604	0.4586

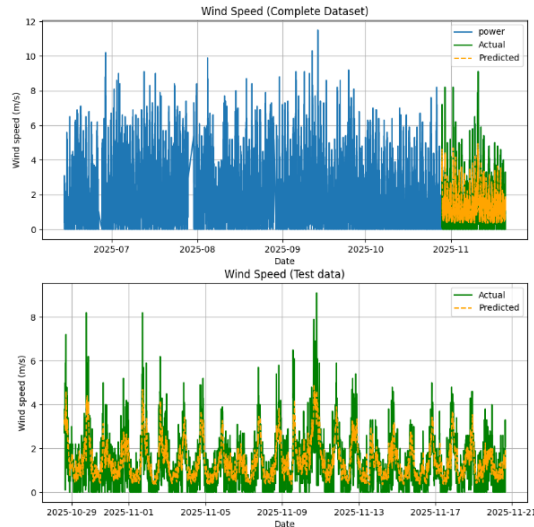


Figure 3. Wind speed prediction from Paiba weather station (Bogotá) using LSTM.

Although wind speed at Anapoima exhibits considerable variability, the models produced an average signal rather than extreme fluctuations, which leads to an underestimation of peak values. An analysis of the R^2 coefficient for both datasets indicates that the predicted values do not represent the observed data, with values of approximately 0.5 for Paiba and 0.3 for Anapoima. Despite the lower prediction error in Anapoima's dataset, a lower R^2 than Paiba's does not imply lower error or better data replication.

The MSTL algorithm was evaluated by varying several parameters, including the forecast horizon and the trend forecaster. The results for the two stations are summarized in Tables 5 and 6. It shows bigger errors than the applied tests to LSTM; this behavior is reproduced across all tests.

Table 3. LSTM error metrics using Relu Function, different Optimizer and Size Window for Anapoima station dataset

Relu		Size window		
Optimizer	Error	10	30	50
Adam	MAE	0.3023	0.3060	0.3031
	MSE	0.1706	0.1715	0.1714
	R^2	0.3816	0.3787	0.3795
	MAE	0.3000	0.2991	0.2961
Nadam	MSE	0.1694	0.1697	0.1686
	R^2	0.3860	0.3852	0.3895
	MAE	0.3038	0.2978	0.2981
Adamax	MSE	0.1685	0.1667	0.1665
	R^2	0.3893	0.3959	0.3970

Table 4. LSTM error metrics using Tanh Function, different Optimizer and Size Window for Anapoima station dataset

Tanh		Size window		
Optimizer	Error	10	30	50
Adam	MAE	0.2998	0.2968	0.3001
	MSE	0.1697	0.1698	0.1714
	R^2	0.3850	0.3848	0.3795
Nadam	MAE	0.29527	0.2955	0.2973
	MSE	0.1704	0.1697	0.1691
	R^2	0.3824	0.3852	0.3876
Adamax	MAE	0.2990	0.3059	0.2968
	MSE	0.1685	0.1717	0.1676
	R^2	0.3893	0.3781	0.3930

The results presented in Table 5 show high errors across all three trend forecasters, even when the horizon is varied. MAE and MSE values are higher than expected and larger than those obtained with the LSTM model. As shown in Figure 4, the predicted signal exhibits a different behavior and fails to reproduce the observed wind speed. Although some spikes are captured corresponding to the daily seasonality, the full signal pattern is not well reproduced, unlike the results obtained with the LSTM. It is important to note that the seasonality of 24 and 144 yielded the lowest error values in the seasonal ARIMA model, while the Simple Exponential Smoothing model consistently showed a reduction in error as the horizon length increased, and the ARCH model yielded the lowest error value with a horizon of 60 (unlike SARIMA) but saw the error rise again when the horizon was extended to 144.

Table 5. MSTL error metrics using different Trend Forecaster and Horizon size with Paiba weather station (Bogotá)

Trend-Forecaster	Error	Horizon		
		24	60	144
Seasonal	MAE	2.4432	3.443	2.3595
ARIMA	MSE	5.9693	12.23	6.7584
Simple	MAE	3.539	2.7276	1.5090
Exponential Smoothing	MSE	12.53	7.7461	3.5683
ARCH	MAE	3.163	1.679	1.811
	MSE	10.007	3.666	4.3163

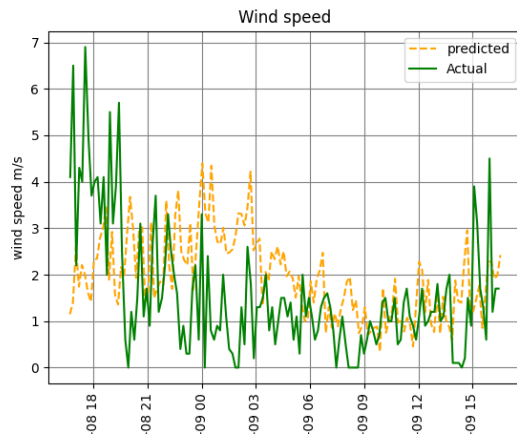
Table 6. MSTL error metrics using different Trend Forecaster and Horizon size with Anapoima weather station data.

Trend-Forecaster	Error	Horizon		
		24	60	144
Seasonal	MAE	0.2490	0.5832	0.3667
ARIMA	MSE	0.0947	0.5561	0.2715
Simple	MAE	0.2590	0.5637	0.3608
Exponential	MSE	0.0955	0.5166	0.2542
Smoothing	MAE	0.4405	0.7483	0.5056
ARCH	MSE	0.3005	0.8337	0.4580

The study showed that predicting wind speed at these two stations using the collected data can be quite challenging, as most tests revealed high errors and low correlation between the actual and measured signals. The various conditions that can affect wind speed, such as temperature and humidity in both cities, can serve as an important benchmark for evaluation. These variables could help to ensure that future predictions better reflect the actual conditions in these areas. It is also important to note that the amount of available data would enable a more targeted analysis and yield more accurate results if a specific period within the year is used.

CONCLUSIONS

The predictions obtained in this study yielded high error values, since wind speed in these locations is highly variable and can be influenced by other factors not considered in this study, such as temperature and humidity.



The figure 4. Wind speed prediction from Paiba weather station (Bogotá) using MSTL.

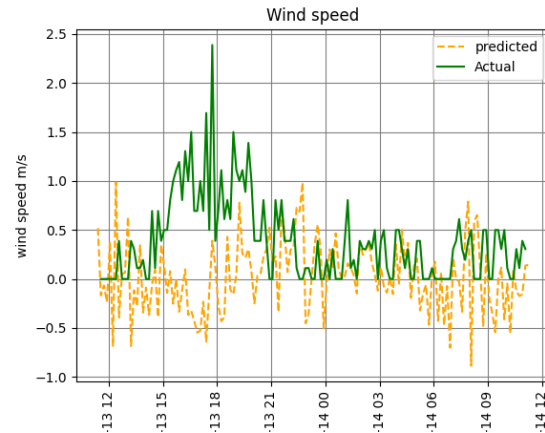


Figure 5. Wind speed prediction from Anapoima weather station using MSTL.

Therefore, the predictions made did not accurately represent the behavior of the study variable; results were found that approximated the true signal's average behavior, and this could lead to significant discrepancies in future applications, since reducing the maximum values in an environment where the greatest amount of energy is extracted precisely from the highest values, rather than from constant values, could lead to these wind-speed-based solutions being discarded, as they would not allow for the proper operation of coupled equipment such as wind turbines. Therefore, it is important to segment the data by a specific time of year, evaluating only the highest and most consistent results, thereby improving prediction during periods with unique characteristics.

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